

Hani Yacoub

Fraud Prevention • AI Abuse • Enforcement Operations @AWS

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PROFESSIONAL SUMMARY

Fraud prevention analyst at AWS. 6+ years in data, risk, and abuse analytics. I build LLM-powered investigation agents, design detection rules across the full account population, and identify how AI services are exploited by fraud actors. I manage precision/recall tradeoffs, build enforcement processes from scratch, and coordinate across engineering, policy, and data science.

CORE SKILLS

Large Language Models & AI Agents · Adversarial Detection & Threat Modeling · Machine Learning (XGBoost, Classifiers) · Enforcement Systems & Policy Design · Python · SQL · Data Pipelines · Cross-Team Coordination & Process Building · Precision/Recall & False-Positive Analysis · Dashboards & Operational Reporting

PROFESSIONAL EXPERIENCE

Fraud Prevention Analyst – AWS Payments & Fraud Prevention

Amazon Web Services (AWS) · Mar 2026 to Present · Berlin, Germany

AI Abuse Detection & Tooling

- First to characterize a novel AI-abuse pattern on an enterprise code-generation service: one payer enrolling thousands of seats on stolen cards to burn model credits, invisible to standard spend controls. Proposed an org-level credit-velocity tripwire.
- Built Claude-based investigation agents that evaluate account signals, automate evidence gathering, and generate enforcement reports. Reduced investigation time from ~1 week to 1 day. Defined output evaluation criteria and operational framework from scratch.
- Built an agent to surface sleeper-account fraud patterns, backed by an XGBoost model (billing spikes, linked accounts, IP-country, payment instruments, risk-score history) with a Streamlit investigation queue.

Enforcement Rules & Systems

- Designed a fraud detection rule using domain age and email-authentication signals. Recall: hundreds of abusive accounts caught. Collateral: near-zero impacts to legitimate customers.
- Added device fingerprint as an ensemble feature to the Fraud Relations model. Result: 34% more disputed dollars caught. Quantified false-positive tradeoffs per signal.
- Traced a multi-six-figure SageMaker abuse case, spanning thousands of jobs and hundreds of thousands of instances, to a regex gap in enforcement code. Identified the root cause and specified the fix for engineering. Deployed an hourly alert to detect recurrence.

Large-Scale Investigations

- Led analysis of a regional fraud-reporting threshold breach. Proved a coordinated ring (not general population fraud) was the driver. Isolated chargebacks to a handful of accounts. Overturned prior assumptions and enabled targeted enforcement.
- Built a fraud-ring detection framework scoring accounts across 13 signals (fingerprint, BIN, ASN, IP, bank, email domain, time zone). 91% of ring members shared 5+ signals. Dispositioned hundreds of accounts with per-account safety carve-outs.
- Built the first fraud-and-revenue dashboard for AWS new EU region. Surfaced fraud-to-revenue ratio and reversal rates for enforcement prioritization.

Data Science & Forecasting

- Delivered a Monte-Carlo human-review volume forecast for a new region's investigation queue (a 40,000-run simulation with p10/p90 planning thresholds and measured coverage). Its central finding inverted the planning assumption: queue volume is driven by operational decisions rather than fraud inflow, and one ring accounted for 71% of all tasks ever queued.

Enforcement Rules & Systems

- Built, deployed, and own a fraud-relations auto-shutdown job running daily in production on AWS's European Sovereign Cloud partition. The eight-gate candidate pipeline routes legitimate-customer carve-outs to review rather than closure, re-checks at enforce time, carries a hard kill switch, and runs a shadow twin that measures precision continuously.

Senior Product Analyst – Risk & Abuse

Zalando · Apr 2024 to Mar 2026 · Berlin, Germany

- Investigated bot networks, account takeovers, and refund abuse across European markets. Translated findings into enforcement policy recommendations and product decisions.
- Developed a 'remaining fraud damage' metric comparing refund rates across segments. Quantified unmitigated fraud and informed defensive investment priorities.
- Measured financial impact of refund and account abuse (Databricks, Azure). Built dashboards consumed by Risk, Product, and Finance leadership.

Data Analyst – Customer Acquisition

AUTO1 Group · Oct 2021 to Apr 2024 · Berlin, Germany

- Built scoring systems to identify high-value B2B customers and churn risk. Enabled the commercial team to prioritize outreach.

Quantitative Analyst Intern – Energy Markets

Statkraft · Jul 2020 to Feb 2021 · Düsseldorf, Germany

PROJECTS

- Renitor:** Built and published Renitor, a VS Code extension (v1.75) for multi-agent AI coding: repo-aware handoffs that carry an unfinished task across 11 coding agents (Claude Code, Codex, Cursor, and more), a multi-model chat with automatic failover when a provider hits its usage cap, and a deterministic safety layer that reviews risky commands before they run. Local-first, zero telemetry. (renitor.com)
- Preflights:** Built Preflights, a pre-import QA tool for Shopify and Google Shopping: a deterministic validation engine that inspects supplier CSVs against live store exports and flags irreversible import risks such as variant deletion, handle collisions, and price and stock conflicts in a safe / review / blocked report. AI is used only to explain findings and suggest fixes, never to decide.

EDUCATION

- Self-directed ML-engineering curriculum, fraud-ML focus (ongoing),** Structured 12-month program: hand-built implementations before libraries, verified against independent implementations at full data scale, 2026 to present
- Data Engineering with AWS Nanodegree,** Udacity, 2024 to 2025
- Master's Degree, Business Management,** SRH Berlin University of Applied Sciences, 2018 to 2021